

Assignment - 1

Q.1) Advantage of DBMS over a file processing system:-

- 1) Data Independence - DBMS provides an abstract view of data, hiding low-level physical details of data representation.
- 2) Efficient data access - DBMS utilizes complex indexing techniques to retrieve data efficiently, drastically reducing I/O costs compared to file systems that must load entire large files.
- 3) Concurrent Access and crash recovery - DBMS schedules ~~concurrent~~ concurrent data accesses safely, ensuring consistency when multiple people access at some time.
- 4) Data Integrity and Security - DBMS can apply integrity constraints and provides access control by offering different customized views for different users.

Not use DBMS when - the application is very simple and the database file size is small, as the high I/O costs and complex app dev. problems in file systems do not arise, making DBMS software overhead significant.

Q.2) Scheme layers:-

- i) Physical (Internal) scheme - Lowest level of abstraction, describes the complete physical details of how data is actually stored and accessed on secondary storage devices, including file formats, indexing, and record formats.

- what data is
- ii) Logical (Conceptual) schema - Middle level, describes ~~only the specific~~ stored in the database (entities, relationships, constraints, etc.) while completely hiding the physical storage details.
- iii) View (External) schema - The highest level, only the specific part of database that a particular user or group is interested in, hiding the rest of the ~~data~~ database to provide customised and ~~auto~~ authorized access.

### Data independence -

- i) Physical - Achieved by the ~~se~~ separation between the physical and logical schemas. ~~for~~ We can change how data is physically stored without rewriting the logical schema.
- ii) Logical - Achieved by the separation b/w logical and view schemas. We can alter the logical structure without affecting the external views of users ~~who~~ who do not rely on that new data.

Q.3) NULL means that a value does not exist or is unknown. It is introduced because -

1. Unknown value - The values exists, but is currently not known.
2. Not Applicable - The attribute does not apply to the specific tuple.

Q.4) The primary design goal for relational databases is schema refinement (normalization) to reduce data redundancy, means storage of duplicate copies of same data.



Goal is desirable as redundancy <sup>causes</sup> several severe problems -

- 1) Redundant storage - Wastes storage space by saving exact same info in ~~not~~ multiple places.
- 2) Update Anomalies - If redundant data is updated in one place but forgotten in another, it leads to data inconsistency.
- 3) Insertion Anomalies - May be impossible to insert new, valid info into database without also being forced to insert unrelated data.
- 4) Deletion Anomalies - Deleting a specific record might unintentionally delete other vital, unrelated info.

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Q.5) i)  $A^+ = \{A, B, C, D, E\} \mid \begin{array}{l} A \rightarrow B, AB \rightarrow C, A \\ C \rightarrow D, D \rightarrow E \end{array}$

$$B^+ = \{B\}$$

$$C^+ = \{C, A, B, D, E\} \mid \begin{array}{l} C \rightarrow A, \dots \end{array}$$

$$D^+ = \{D, E\} \mid E^+ = \{E\}$$

So,  $\{A\}, \{C\}$  candidate keys.

ii)  $B^+ = \{B, E, A, D, C\} \mid \begin{array}{l} B \rightarrow E, E \rightarrow A, A \rightarrow D, D \rightarrow C \end{array}$

$$A^+ = \{A, D, C\}$$

$$C^+ = \{C\}, D^+ = \{D, C\}$$

$$E^+ = \{E, A, D, C\}$$

So, only  $\{B\}$  as candidate key.

iii) Checking single attributes first ✓

$$C^+ = \{C, D, E, B, A\}$$

$$C \rightarrow D, D \rightarrow E, D \rightarrow B, E \rightarrow A$$

$$D^+ = \{D, B, E, A, C\}$$

$$D \rightarrow E, D \rightarrow B, E \rightarrow A, AB \rightarrow C$$

$$E^+ = \{E, A\}, A^+ = \{A\}, B^+ = \{B\}$$

Now pairs,  $(AB)^+ = \{A, B, C, D, E\}$

$$AB \rightarrow C, D \rightarrow E$$

$$(BE)^+ = \{B, E, A, C, D\}$$

$$C \rightarrow D, D \rightarrow E, E \rightarrow A, AB \rightarrow C, C \rightarrow D$$

only these,

so,  ~~$\{C\}, \{D\}, \{A, B\}, \{B, E\}$~~  candidate keys.

iv) A B not in RHS of any, it must be in every candidate key.

$$B^+ = \{B\} \quad \times$$

now pairs,

$$(AB)^+ = \{A, B, C, D, E, F\}$$

$$AB \rightarrow C, C \rightarrow DE, E \rightarrow F$$

$$(BC)^+ = \{B, C, D, E, F, A\}$$

$$C \rightarrow DE, E \rightarrow F, F \rightarrow A$$

$$(BD)^+ = \{B, D\}$$

$$(BE)^+ = \{B, E, F, A, C, D\}$$

$$E \rightarrow F, F \rightarrow A, AB \rightarrow C, E \rightarrow DE$$

$$(BF)^+ = \{B, F, A, C, D, E\}$$

$$F \rightarrow A, AB \rightarrow C, C \rightarrow DE$$

so,  ~~$\{A\}, \{B\}, \{C\}$~~

$\{A, B\}, \{B, C\}, \{B, E\}, \{B, F\}$

Candidate keys.

$$v) A^+ = \{A, B, C, D, E, G, F, H\}$$

$$A \rightarrow B, AB \rightarrow CD, C \rightarrow EF, D \rightarrow EG, F \rightarrow H$$

$$C^+ = \{C, E, F, A, H, B, D, G\}$$

$$C \rightarrow EF, F \rightarrow H, H \rightarrow A, \dots$$

$$F^+ = \{F, H, A, B, C, D, E, G\}$$

$$F \rightarrow H, H \rightarrow A, \dots$$

$$H^+ = \{H, A, B, C, D, E, G, F\}$$

$$H \rightarrow A, \dots$$



So,  $\{A\}$ ,  $\{C\}$ ,  $\{F\}$ ,  $\{H\}$  are candidate keys.

Q.6]

i)  $D_1 = \{ \underset{R_1}{ABC}, \underset{R_2}{ADE}, \underset{R_3}{BF}, \underset{R_4}{FGH}, \underset{R_5}{DIJ} \}$

$R_1 \cap R_2 = \{A\}$ , and  $A \rightarrow DE$ , so  $A$  uniquely identifies  $A, D, E$

$\Rightarrow$  join is lossless,  $R_{12}(ABCDE)$ .

$R_3 \cap R_4 = \{F\}$ , and  $F \rightarrow GH$ , identifies  $FGH$   
Hence ~~is~~ lossless,  $R_{34}(BFGH)$ .

$R_{12} \cap R_5 = \{D\}$ , as  $D \rightarrow IJ$ , identifies  $DIJ$   
So, lossless,  $R_{125}(ABCDEIJ)$

$R_{125} \cap R_{34} = \{B\}$ , as  $B \rightarrow F$ ,  $F \rightarrow GH$   
So, identifies  $B, F, G, H$

Hence, lossless  $\Rightarrow R(ABCDEFGH IJ)$

4th lossless decomposition.

ii)  $D_2 = \{ \underset{R_1}{ABCDE}, \underset{R_2}{BFGH}, \underset{R_3}{DIJ} \}$

$R_1 \cap R_2 = \{D\}$ , and  $D$  identifies  $DIJ$  (from i)  
So, lossless and  $R_{12} = \{ABCDEIJ\}$

$R_{12} \cap R_3 = \{B\}$ , and  $B$  identifies  $BFGH$  (from i)  
So, lossless  $\Rightarrow R(ABCDEFGH IJ)$

4th lossless decomposition.

$$iii) D_3 = \{ \underset{R_1}{ABCD}, \underset{R_2}{DE}, \underset{R_3}{BF}, \underset{R_4}{FGH}, \underset{R_5}{DIJ} \}$$

Let's join  $R_1, R_2$  -  
 $R_1 \cap R_2 = \{D\} \mid D^+ = \{D, I, J\}$

So,  $D^+$  doesn't contain E (can't determine DE)  
 neither A, B, C (can't determine ABCD)

Hence, Lossy decomposition.

Q.7] i)  $D_1 = \{ABC, ACDE, ADG\}$

Local FD dependencies (FD) for each scheme  $\Rightarrow$

(F<sub>1</sub>)  $ABC = \{AB \rightarrow C, AC \rightarrow B, BC \rightarrow A\}$

(F<sub>2</sub>)  $ACDE = \{AD \rightarrow E\}$

(F<sub>3</sub>)  $ADG = \{\emptyset\}$

So, FDs  $B \rightarrow D$  and  $E \rightarrow G$  are broken

test if missing FDs can be find using only local dependencies ( $F_1 \cup F_2 \cup F_3$ )

$B^+ = \{B\}$ , as we can't do  $AB \rightarrow C, CB \rightarrow A$

So, D is not in closure of B, dependency permanently lost.

$\Rightarrow$  Not dependency preserving.

ii)  $D_2 = \{ABC, ADEG\}$

(F<sub>1</sub>)  $ABC = \{AB \rightarrow C, BC \rightarrow A, AC \rightarrow B\}$

(F<sub>2</sub>)  $ADEG = \{AD \rightarrow E, E \rightarrow G\}$

Now, FD  $B \rightarrow D$  is missing, so check in  $F_1 \cup F_2$

$B^+ = \{B\}$ , again  $B \rightarrow D$  is lost.

$\Rightarrow$  Not dependency preserving.